

14.7 Solution: (a) Since the particle is assumed to move in a straight line with constant speed, we can write

$$dt = \frac{dx}{v_0}.$$

Then,

$$P dt = \frac{2}{3} \frac{z^2 e^2}{m^2 c^3} |\nabla V|^2 \frac{dx}{v_0} = \frac{2}{3} \frac{z^2 e^2}{m^2 c^3} \frac{z^2 Z^2 e^4}{(x^2 + b^2)^2} \frac{dx}{v_0}.$$

Therefore, the total energy radiated is

$$\Delta W = \frac{2}{3} \frac{z^4 Z^2 e^6}{m^2 c^3 v_0} \int_{-\infty}^{+\infty} \frac{dx}{(x^2 + b^2)^2} = \frac{2}{3} \frac{z^4 Z^2 e^6}{m^2 c^3 v_0} \frac{\pi}{2b^3} = \frac{\pi z^4 Z^2 e^6}{3m^2 c^3 v_0} \frac{1}{b^3}.$$

(b) For

$$b = r_c = \frac{2zZe^2}{mv_0^2},$$

we have

$$\Delta W = \frac{\pi}{24} \frac{zm v_0^5}{Zc^3},$$

which is comparable to Porb. 14.5 (b).

(c) The radiation cross section can be calculated as

$$\chi = 2\pi \int_{b_{\min}}^{+\infty} \Delta W(b) b db = \frac{2\pi^2 z^4 Z^2 e^6}{3m^2 c^3 v_0} \int_{b_{\min}}^{+\infty} \frac{db}{b^2} = \frac{2\pi^2 z^4 Z^2 e^6}{3m^2 c^3 v_0 b_{\min}}.$$

Following Eq. (13.16), we can set

$$b_{\min} = \frac{\hbar}{mv_0},$$

then

$$\chi = \frac{2\pi^2}{3} Z \left(\frac{Ze^2}{\hbar c} \right) \frac{z^4 e^4}{mc^2},$$

which is close to the Heitler-Bethe result, with $2\pi^2 \approx 19.739$.